

 UNIVERSITY OF TECHNOLOGY - VNUHCM FACULTY OF COMPUTER SCIENCE AND ENGINEERING	HOMEWORK 1		Semester/Academic year		1	2026-2027
			Due		07/10/2026	
	Course title	Machine Learning				
	Course ID	CO3117				
	Duration	Take-home	Homework set		HW1	
Notes: Handwritten homework, submitted as homework/hw1-submission.pdf with tag hw1.						

Instructions

- **Foundations: workflow, evaluation and generalisation** (Chapter 1). Five problems, each worth 2 points (0.5 per part); total 10 points.
- **Write by hand**, on paper or with a stylus. Show your working: a bare number earns no credit. Put your full name, student ID and the set number at the top of the first page.
- **Work on your own**. Do not use AI tools, solution manuals or classmates' work before your submission is tagged. You may use your lecture notes and the course slides. Any later AI use goes in AI_USE.md, as for the weekly drills.
- **Submit in your course repository**. Scan all pages, in order, into one PDF of at most 5 MB, named homework/hw1-submission.pdf. Commit it, run python tools/check.py --checkpoint hw1, then git tag hw1 and git push origin main hw1.
- **Deadline: 07/10/2026 07:00** (the time GitHub receives the tag hw1). Solutions are released on 07/10/2026. Afterwards you may add homework/hw1-corrections.md; never replace or re-tag the submission.

Problem 1 (L.O.1) (2 pts): Answer the following questions about the definition and workflow of machine learning:

- A. State Mitchell's definition of machine learning in terms of a task T , a performance measure P and experience E .
- B. A post office wants a system that reads handwritten postal codes on scanned envelopes. Identify T , P and E for this system.
- C. How does building this system with machine learning differ from building it with traditional, rule-based programming?
- D. List, in order, the phases of a machine learning project from problem definition to after deployment.

Problem 2 (L.O.1) (2 pts): Answer the following questions about categories of models and about evaluation:

- A. Distinguish parametric from non-parametric models, with one example of each.
- B. Distinguish generative from discriminative models: which distribution does each learn? Give one example of each.
- C. A regression model predicts $\hat{y} = (2, 5, 4, 7)$ for targets $y = (3, 5, 2, 8)$. Compute MAE, MSE and RMSE.
- D. In bootstrap evaluation, a sample of size n is drawn with replacement from n examples. What is the probability that a given example is never drawn, and what is its limit for large n ? How are these examples used?

Problem 3 (L.O.1) (2 pts): Answer the following questions about model evaluation:

- A. Explain the roles of the training set, the validation set and the test set.
- B. Describe k -fold cross-validation. For 500 labelled examples and $k=5$, how many models are trained, and how many examples are used for training and for testing in each fold?
- C. Why must the test set not be used to choose hyperparameters?
- D. A team has only 60 labelled examples. Why is a single 80/20 hold-out split an unreliable estimate of performance here, and what extra information does 5-fold cross-validation provide?

Problem 4 (L.O.3) (2 pts): A binary classifier for detecting fraudulent transactions is tested on 200 transactions. Its confusion matrix is: TP = 40, FN = 10, FP = 20, TN = 130, where “positive” means fraud. Answer the following questions:

- A. Compute accuracy, precision, recall and F1-score for the positive class.
- B. A trivial model that labels every transaction as not fraud is proposed. Compute its accuracy and recall, and explain why accuracy alone is misleading here.
- C. For fraud detection, which error is usually more costly, a false negative or a false positive? Which metric should therefore be monitored closely?
- D. Define the ROC curve and the AUC. What does an AUC of 0.5 mean, and how can AUC be interpreted as a probability?

Problem 5 (L.O.1) (2 pts): Answer the following questions about underfitting, overfitting and the bias-variance trade-off:

- A. Three models have the following errors. Model P: training 25%, test 27%. Model Q: training 2%, test 24%. Model R: training 8%, test 10%. Classify each model as underfitting, overfitting or a good fit.
- B. Write the bias-variance decomposition of the expected squared error of a regression model and explain each term.
- C. How do bias and variance change as model complexity increases? Sketch or describe the typical training and test error curves.
- D. An estimator has bias 0.3 and variance 0.05 at a point, and the noise variance is 0.1. Compute its expected squared error at that point. Which term cannot be reduced by any model?

--- END ---